T3000 API & Protocol

Widget Types · built Nov 13, 2026

Widget Types

All widget types available for LCD page definitions. Each maps to an LVGL equivalent on the device firmware.

Widget Type Reference

`large_text` — Big Number Display

Used for prominent temperature readings. Spans multiple grid rows.

JSON

```
{
  "type": "large_text",
  "field": "temp",
  "row": 1,
  "col": 0,
  "colSpan": 17,
  "rowSpan": 3,
  "fontSize": "72px",
  "suffix": "°C",
  "align": "center"
}
```

Property	Type	Required	Description
`field`	string	Yes	Data field to display (e.g., `temp`, `stp`)
`row`	number	Yes	Grid row (0-based)
`col`	number	No	Grid column (0-based), default 0
`colSpan`	number	No	Number of columns to span, default 17 (full width)
`rowSpan`	number	No	Number of rows to span, default 2
`fontSize`	string	No	CSS font size, default "72px"
`suffix`	string	No	Text after value (e.g., "°C", "%")
`align`	string	No	"center", "left", "right"

LVGL equivalent: `lv_label` with large font

`label_value` — Label + Value Row

The most common widget. Shows a label on the left and a value in a rounded box on the right. Used for editable settings rows.

JSON

```
{
  "type": "label_value",
  "id": "stp",
  "label": "STP",
  "field": "stp",
  "register": 345,
  "row": 3,
  "editable": true,
  "options": [10, 15, 20, 25, 30],
  "maxValue": 35,
  "minValue": 5,
  "suffix": "°C"
}
```

Property	Type	Required	Description
`id`	string	Yes	Unique widget identifier
`label`	string	Yes	Left-side text label
`field`	string	Yes	Data field for the value
`register`	number	No	Modbus register number for sync
`row`	number	Yes	Grid row
`editable`	boolean	No	Can user change this value? Default false
`options`	array	No	Allowed values (cycle through with ▲/▼)
`maxValue`	number	No	Max for numeric increment
`minValue`	number	No	Min for numeric increment
`suffix`	string	No	Unit suffix
`navigateTo`	string	No	Page ID to navigate to when selected

LVGL equivalent: `lv_obj` container with `lv_label` (label) + `lv_label` (value in box)

`edit_value` — Standalone Editable Value

A large editable number in a rounded box. Used on dedicated edit pages (e.g., Setpoint page).

JSON

```
{
  "type": "edit_value",
  "field": "stp",
  "register": 345,
  "row": 2,
  "fontSize": "36px",
  "editable": true,
  "minValue": 5.0,
  "maxValue": 35.0,
  "step": 0.5
}
```

Property	Type	Required	Description
`field`	string	Yes	Data field
`register`	number	No	Modbus register
`row`	number	Yes	Grid row
`fontSize`	string	No	Default "36px"
`editable`	boolean	No	Default true
`minValue`	number	No	Minimum allowed
`maxValue`	number	No	Maximum allowed
`step`	number	No	Increment step (e.g., 0.5 for temperature)

LVGL equivalent: `lv_label` with `lv_obj` border/background

`header` — Page Title

Large centered text at the top of a page.

JSON

```
{
  "type": "header",
  "text": "Communication\nSettings",
  "row": 1,
  "fontSize": "24px"
}
```

Property	Type	Required	Description
`text`	string	Yes	Title text (supports `\n` for line breaks)
`row`	number	No	Starting grid row, default 1
`fontSize`	string	No	Default "24px"

LVGL equivalent: `lv_label` centered

`text` — Static or Data-Bound Text

General-purpose text element. Can be static ("Inside") or data-bound (IP address).

JSON

```
{
  "type": "text",
  "text": "Inside",
  "row": 4,
  "col": 8,
  "fontSize": "18px"
}
```

Or data-bound:

JSON

```
{
  "type": "text",
  "field": "ip",
  "row": 6,
  "col": 0,
  "fontSize": "18px"
}
```

Property	Type	Required	Description
`text`	string	No	Static text (use this OR `field`, not both)
`field`	string	No	Data field for dynamic text
`row`	number	Yes	Grid row
`col`	number	No	Grid column, default 0
`fontSize`	string	No	Default matches page style
`suffix`	string	No	Appended to value (e.g., "%")
`color`	string	No	Text color override

LVGL equivalent: `lv_label`

`icon` — Single Icon

Displays a named icon at a grid position.

```JSON
{
  "type": "icon",
  "icon": "house",
  "row": 4,
  "col": 1,
  "size": "32px"
}
```

Property	Type	Required	Description
`icon`	string	Yes	Icon name (see Icon Library below)
`row`	number	Yes	Grid row
`col`	number	Yes	Grid column
`size`	string	No	Icon size, default "32px"
`opacity`	number	No	0.0 to 1.0, for dimmed/active states

Available icons: `house` , `moon` , `snowflake` , `fan` , `wifi` , `sun` , `leaf` , `thermometer`

LVGL equivalent: `lv_img` or custom bitmap

`icon_bar` — Row of Navigation Icons

A horizontal row of icons at the bottom of the page, used for page navigation.

JSON

```
{
  "type": "icon_bar",
  "row": 9,
  "icons": [
    { "icon": "moon", "pageId": "schedule" },
    { "icon": "house", "pageId": "main" },
    { "icon": "snowflake", "pageId": "inside" },
    { "icon": "fan", "pageId": "fan_control" }
  ]
}
```

Property	Type	Required	Description
`row`	number	Yes	Grid row
`icons`	array	Yes	Array of icon objects
`icons[].icon`	string	Yes	Icon name
`icons[].pageId`	string	No	Page to navigate to when selected

LVGL equivalent: Row of `lv_img` buttons

`wifi_icon` — WiFi Signal Indicator

Special widget for WiFi signal strength, always top-right.

JSON

```
{
  "type": "wifi_icon",
  "row": 0,
  "col": 16
}
```

Property	Type	Required	Description
`row`	number	No	Default 0
`col`	number	No	Default 16 (far right)

LVGL equivalent: `lv_img` with signal strength variants

`divider` — Horizontal Line

A separator line spanning the full width.

JSON

```
{
  "type": "divider",
  "row": 3,
  "color": "#ffffff",
  "opacity": 0.3
}
```

Property	Type	Required	Description
`row`	number	Yes	Grid row to place the line
`color`	string	No	Line color, default "#ffffff"
`opacity`	number	No	Default 0.3

LVGL equivalent: `lv_line`

`footer_hint` — Hint Text

Small text at the bottom showing button hints.

JSON

```
{
  "type": "footer_hint",
  "text": "+  Edit  -",
  "row": 8
}
```

LVGL equivalent: `lv_label` at bottom area

`footer_nav` — Navigation Text

Navigation hints at the very bottom.

```JSON
{
  "type": "footer_nav",
  "text": "< Back   Next >",
  "row": 9
}
```

LVGL equivalent: `lv_label` at bottom area

Widget Type Summary

Type	Purpose	Editable	LVGL
`large_text`	Big temperature/number display	No	`lv_label`
`label_value`	Settings row (label + value)	Yes	`lv_obj` + `lv_label` × 2
`edit_value`	Standalone editable number	Yes	`lv_label` + `lv_obj`
`header`	Page title	No	`lv_label`
`text`	Static or dynamic text	No	`lv_label`
`icon`	Single icon	No	`lv_img`
`icon_bar`	Navigation icon row	No	`lv_img` × N
`wifi_icon`	WiFi signal	No	`lv_img`
`divider`	Horizontal line	No	`lv_line`
`footer_hint`	Button hints	No	`lv_label`
`footer_nav`	Navigation hints	No	`lv_label`